

IP6: a novel anti-cancer agent.

Inositol hexaphosphate (InsP6 or IP6) is ubiquitous. At 10 microM to 1 mM concentrations, IP6 and its lower phosphorylated forms (IP(1-5)) as well as inositol (Ins) are contained in most mammalian cells, wherein they are important in regulating vital cellular functions such as signal transduction, cell proliferation and differentiation. A striking anti-cancer action of IP6 has been demonstrated both in vivo and in vitro, which is based on the hypotheses that exogenously administered IP6 may be internalized, dephosphorylated to IP(1-5), and inhibit cell growth. There is additional evidence that Ins alone may further enhance the anti-cancer effect of IP6. Besides decreasing cellular proliferation, IP6 also causes differentiation of malignant cells often resulting in a reversion to normal phenotype. These data strongly point towards the involvement of signal transduction pathways, cell cycle regulatory genes, differentiation genes, oncogenes and perhaps, tumor suppressor genes in bringing about the observed anti-neoplastic action of IP6.